

Unsupervised Anomaly Detection on SMD: Autoencoder vs. Zero-Shot MOMENT

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Abstract

This report compares two reconstruction-based approaches for unsupervised anomaly detection on the SMD machine-1-1 sequence. The training data is treated as normal, and test labels are used only for final evaluation. I compare a convolutional autoencoder trained on normal windows with MOMENT, a pre-trained time-series foundation model used zero-shot. To keep the comparison fair, both methods use the same preprocessing, window length, anomaly score and calibration-based threshold. The results show that both models produce informative anomaly scores, but MOMENT gives a much cleaner binary detector after thresholding, reaching a window-level F1 of 0.8190 against 0.3612 for the autoencoder.

1 Problem Setup

I use the Server Machine Dataset (SMD) machine-1-1 sequence from the OmniAnomaly benchmark [1]. The train and test arrays both have shape 28479×38 : each timestamp is described by 38 server measurements. The test labels are provided at timestamp level, with 0 for normal and 1 for anomalous behavior. In this machine, the test set contains 2694 anomalous timestamps, i.e. a timestamp-level anomaly rate of 9.46%.

The first point of the protocol is to avoid using information from the test set before evaluation. For each channel j , I compute the mean and standard deviation on the normal training sequence only, then standardize both train and test with these training statistics:

$$\tilde{x}_{t,j} = \frac{x_{t,j} - \mu_j^{\text{train}}}{\sigma_j^{\text{train}}}.$$

Channels with zero variance in the training sequence are handled by setting $\sigma_j^{\text{train}} = 1$. This keeps the preprocessing well-defined while preserving the fact that these channels are constant on the normal training data.

Both methods operate on windows rather than on individual timestamps. I use windows of length $L = 96$ for the two models. This is compatible with MOMENT reconstruction mode and, more importantly, keeps the comparison fair: the autoencoder and MOMENT see the same temporal context. A test window starting at s is assigned the label

$$y_s^{\text{win}} = \max_{t=s}^{s+L-1} y_t,$$

so a window is anomalous if it contains at least one anomalous timestamp. This creates 28384 test windows. The window anomaly rate is 12.17%, higher than the timestamp anomaly rate because overlapping windows around an anomalous timestamp also become positive.

For the autoencoder, the normal training windows are split chronologically. The first 80% are used to fit the model, and the last 20% are kept as a normal calibration segment. This matters because the threshold should not be estimated on exactly the same windows used for gradient updates. In the final notebook, stride 1 gives 22688 fitting windows and 5601 calibration windows. MOMENT is not trained on SMD, but I use the same calibration windows to set its threshold, so both methods use the same thresholding protocol.

2 Models and Scores

Autoencoder. The autoencoder receives tensors of shape 38×96 . The encoder uses one-dimensional convolutions with channel sizes $38 \rightarrow 64 \rightarrow 32 \rightarrow 16$, followed by a symmetric decoder $16 \rightarrow 32 \rightarrow 64 \rightarrow 38$. It is trained for 50 epochs with Adam, learning rate 10^{-3} , and mean squared reconstruction loss. Since it only sees normal fitting windows, the intended behavior is simple: normal windows should be reconstructed with small error, while windows that do not look like the normal training regime should have larger error.

MOMENT zero-shot. MOMENT [2] is loaded as MOMENT-1-large in reconstruction mode. Its parameters are frozen; there is no fine-tuning on SMD. This is the zero-shot foundation-model part of the project. The only thing extracted from MOMENT is its reconstruction of the input window.

For both methods, I use the same anomaly score. If $x \in \mathbb{R}^{38 \times L}$ is a standardized window and $\hat{x}$ its reconstruction, then

$$s(x) = \frac{1}{38L} \sum_{j=1}^{38} \sum_{t=1}^L (x_{j,t} - \hat{x}_{j,t})^2.$$

For the autoencoder, this is exactly the loss minimized during training. For MOMENT, the same score is natural because the model is used as a reconstruction model: if a window follows regular temporal structure, the reconstruction should be easier; if it is abnormal, the reconstruction error should tend to increase. No anomaly labels are used to define this score.

3 Thresholding and Evaluation

AUROC and AUPRC are reported first because they evaluate anomaly scores without fixing a threshold: AUROC measures ranking quality, while AUPRC focuses on the precision-recall trade-off.

Since SMD annotations are timestamp-level while the models score windows, I report two threshold-free evaluations. Window-level AUROC/AUPRC uses the labels y_s^{win} . Timestamp-level AUROC/AUPRC is obtained by mapping overlapping window scores back to timestamps:

$$\bar{s}_t = \frac{1}{|\mathcal{W}(t)|} \sum_{s \in \mathcal{W}(t)} s(x_s),$$

where $\mathcal{W}(t)$ is the set of windows that contain timestamp t .

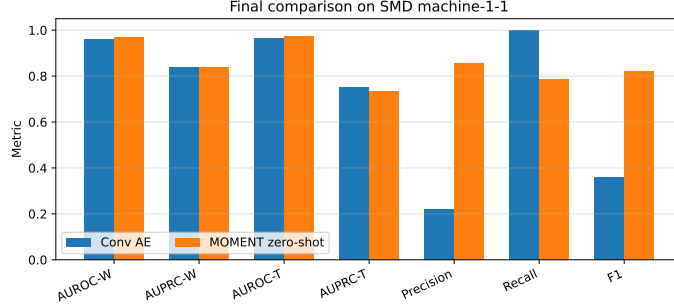
For binary metrics, a threshold is needed: precision measures alert quality, recall measures anomaly coverage, and F1 summarizes their trade-off. The threshold is chosen only from the held-out normal calibration scores, before looking at the test labels. Let $s_1^{\text{cal}}, \dots, s_n^{\text{cal}}$ be the normal calibration scores, and let m be the number of test windows. I use a false-alarm budget $\delta = 0.05$ over the full test horizon.

If, under normal behavior, a single test window exceeds the threshold with probability at most α , then by the union bound

$$\mathbb{P}(\exists i \in \{1, \dots, m\} : s_i > \tau) \leq \sum_{i=1}^m \mathbb{P}(s_i > \tau) \leq m\alpha.$$

Table 1: Final anomaly detection results on SMD machine-1-1.

Method	AUROC-W	AUPRC-W	AUROC-T	AUPRC-T	Prec.	Rec.	F1	FP	FN	τ
Conv. AE	0.9596	0.8366	0.9656	0.7508	0.2204	1.0000	0.3612	12,215	0	0.0625
MOMENT	0.9697	0.8391	0.9739	0.7317	0.8556	0.7855	0.8190	458	741	1.7406

**Figure 1:** Final comparison between the autoencoder and zero-shot MOMENT.

Choosing $\alpha = \frac{\delta}{m}$ therefore gives

$$\mathbb{P}(\text{at least one false alarm over the test horizon}) \leq \delta.$$

Here $m = 28384$, so

$$\alpha = \frac{0.05}{28384} \simeq 1.76 \times 10^{-6}.$$

The target threshold is thus the $(1 - \alpha)$ -quantile of the normal score distribution:

$$\tau = F_0^{-1}(1 - \alpha),$$

where F_0 is the distribution of anomaly scores under normal behavior.

Since F_0 is unknown, it is estimated from the held-out calibration scores. However, with only $n = 5601$ normal calibration windows, this extreme quantile cannot be estimated finely. The empirical rule therefore becomes

$$\tau = \max_{1 \leq i \leq n} s_i^{\text{cal}}.$$

This is a conservative split-calibration rule: the threshold is fixed before seeing the test labels, but the finite calibration sample means that the bound should be read as a target false-alarm budget, not as an exact guarantee on the observed test false-positive rate.

Threshold sensitivity curves are shown only as diagnostics. They help interpret the precision–recall trade-off, but are not used to choose the reported threshold.

4 Results

The threshold-free metrics show that both reconstruction scores separate anomalous from normal behavior well. MOMENT obtains the best window-level AUROC and AUPRC, as well as the best timestamp-level AUROC, while the autoencoder has a slightly higher timestamp-level AUPRC. This suggests that both scores are informative, with the autoencoder being particularly sensitive around anomalous regions.

The main difference appears after applying the fixed calibration threshold. The autoencoder reaches recall 1.0000, detecting all anomalous windows, but at the cost of 12215 false positives,

precision 0.2204, and F1 0.3612. It is therefore very sensitive, but not selective.

MOMENT has lower recall (0.7855), missing 741 anomalous windows, but produces only 458 false positives. This gives precision 0.8556 and F1 0.8190, making it a much more usable binary detector under the same calibration rule. This suggests that MOMENT’s pretrained temporal prior gives more stable reconstruction errors on normal test windows, whereas the local autoencoder overreacts to normal distribution shifts.

This comparison is not perfectly symmetric: the autoencoder is trained only on the normal sequence of one SMD machine, while MOMENT benefits from large-scale time-series pretraining. This is precisely the point of the experiment. On machine-1-1, the zero-shot foundation model provides a stronger reconstruction prior for binary anomaly detection under the fixed calibration protocol.

5 Conclusion

This experiment shows that reconstruction-based anomaly detection can work well on SMD, but that the quality of the reconstruction prior matters as much as the scoring rule itself. Under the same preprocessing, windowing, score definition and calibration protocol, the local autoencoder detects nearly all anomalous windows but produces many false alerts. In contrast, zero-shot MOMENT gives a more selective detector, with far fewer false positives and a much higher window-level F1 score. This suggests that, on this machine, the pretrained temporal representation of MOMENT provides a stronger notion of normality than a small autoencoder trained only on the available normal sequence.

The code used to reproduce these experiments is available at: <https://github.com/medelbechir/Deep-Learning-for-Time-Series>.

References

- [1] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei. Robust anomaly detection for multivariate time series through stochastic recurrent neural network. *KDD*, 2019.
- [2] M. Goswami et al. MOMENT: A family of open time-series foundation models. 2024.